Sepehr Saeedpour

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Education

- **École Normale Supérieure (ENS) - PSL University** — Paris, France
  Normalien and M.Sc. in Cognitive Science (Modeling, Neurotheory and AI Track) — *2024 - now*
  - Thesis: Inferring Head-Direction Ring Attractor Dynamics with Gaussian-Process State Space Models
- **University of Tehran - School of Electrical and Computer Engineering** — Tehran, Iran
  B.Sc. in Electrical Engineering — *2018 - 2023*
  - Thesis: Interindividual differences in Learning to Act or Avoid.
  - Awarded Best Undergraduate Thesis in the ECE School.

Research Experience

- **Sorbonne université — Institut des Systèmes Intelligents et de Robotique (ISIR)** — Paris, France
  Research Intern — *Aug 2025 – Present*
  Supervisor: Dr. Heike Stein
  - Modeling population-level neural dynamics in the head-direction system using Gaussian-process state-space models (GPSSMs), treating observed spiking activity as noisy evidence about a low-dimensional latent angular state evolving under nonlinear dynamics.
- **École Normale Supérieure (ENS) — Human Reinforcement Learning Lab** — Paris, France
  Research Intern — *Feb 2024 – June 2025*
  Co-Supervisor: Dr. Stefano Palminteri, Dr. Hernan Anllo
  - Investigated the computational mechanisms of experiential knowledge transmission (e.g., teaching) by designing and implementing an online behavioural experiment (demo), recruiting and testing 50 participants.
- **Ludwig Maximilians Universität München - Crowd Cognition Group** — Remote
  Undergraduate Research Assistant — *June 2021 - August 2024*
  Co-supervisors: Dr. Bahador Bahrami, Prof. Dr. Ophelia Deroy
  - Fitted reinforcement-learning models to behavioral data from a Go/NoGo task to infer latent cognitive parameters underlying Pavlovian–instrumental interactions; published results as first author.
  - Quantified goal-directed vs. habitual control in social decision-making by fitting model-based and model-free reinforcement-learning models to behavioral data and co-authoring the manuscript.
  - Developed real-time eye-tracking pipelines for interactive environments and characterized gaze dynamics using time-series analysis and visual saliency representations.

Publications

- Zhong, Z., Prystawski, B., Wu, S. A., Fascendini, B., **Saeedpour, S.**, Austerweil, J. L. (2026). Interpretational alignment: How agents learn from physical guidance depends on how they interpret it. Accepted in CogSci Proceedings Conference 2026.
- **Saeedpour, S.**, Miandari, M., Deroy, O., Bahrami, B. (2023). Interindividual differences in Pavlovian influence on learning are consistent, Royal Society Open Science, doi.org/10.1098/rsos.230447
- Navidi, P., **Saeedpour, S.**, Miandari, M., Ershadmanesh, S., Bahrami, B. (2023). Prosocial learning: Model-based or model-free?, PLoS One, doi.org/10.1371/journal.pone.0287563

Poster Presentations

- **Inferring Ring-Attractor Dynamics in Head-Direction Neural Activity.** Presented at the PINTS (Paris Neuroscience, Theory and Systems) Conference, Paris, Fall 2025.
- **Teaching by Description Can Transfer Rule Policies in a Deterministic Contextual Bandit.** Presented at the COSMOS (Computational Summer school on Modeling Social and collective behavior) Summer School, Tokyo, Japan, Summer 2025.

Honors, Awards and Scholarships

- **COSYNE Mentorship Travel Grant** 2026
  Computational and Systems Neuroscience Conference.
- **Admitted to the Concours Normalien Étudiant.** École Normale Supérieure-PSL, Cognitive Science 2024
  Highly selective french competitive exam granting Normalien student status at ENS.
- **Selected and funded for COSMOS Summer School** 2025
  COSMOS (Computational Summer school on Modeling Social and collective behavior) Summer School, Tokyo, Japan, Summer 2025.
- **The Best Undergraduate Thesis Award.** ECE school, University of Tehran. (Link) 2022
- **International Collegiate Programming Contest (ICPC) world finalist.** Bangladesh. (Link) Nov 2022
  Coached the team "aFunnyRandomPhrase" from the Univesity of Tehran
- **Silver Medalist** in the 12th National Olympiad of Astronomy and Astrophysics (NOAA) 2016

Work Experience

- **YUZ Games Studio - Cafe Bazaar** Tehran, Iran
  Game Data Analyst *Apr. 2021 - Aug. 2022*
  - Employed statistical and machine learning approaches to analyze YUZ games' users' behavior in the game and optimizing various aspects of the game and levels to improve gamers' experience.
- **People Analytics Team - Cafe Bazaar** Tehran, Iran
  Human Resources Data Analyst *Jun. 2019 - Apr. 2021*
  - Utilized data-driven approaches to analyze employees' profiles to optimise work conditions.

Teaching Experience

- **Teaching Assistant - University of Tehran**
  - **Machine Learning**: Instructors: Prof. Nadjar Araabi & Dr. Abolghasemi Dahaqani Fall 2020
  - **Engineering Probability and Statistics**: Instructor: Dr. Behnam Bahrak Spring 2020
  - **Engineering Probability and Statistics**: Instructor: Dr. Abolghasemi Dahaqani Fall 2019
- **Instructor, Astronomy & Astrophysics Olympiad** 2017 - 2019
  Allame Helli 1 High school
  - Taught Data Analysis methods (e.g. linear models, clustering) to students preparing for the National Olympiad in Astronomy & Astrophysics (NOAA).

Organizational Experience

- **Executive Director at CNCS 2019** Winter 2019
  University of Tehran (Cognitive Neuroscience Competiton for Students)
  - Managed a team to execute the event.
  - Gathered speakers on different topics in cognitive science.
- **Performing Arts Club Vice Chair** 2019
  University of Tehran
  - Managed the activities of Theater groups, shows and performances, workshops and events around Performing Arts in the University of Tehran.